AP PRECALCULUS · TOPIC 1.6

Polynomial Functions and End Behavior

Section 1 — Multiple Choice
Circle the letter of the correct answer

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

1 Let $f(x) = -2x^5 + 3x^2 - 7$. Which statement describes the end behavior of f?

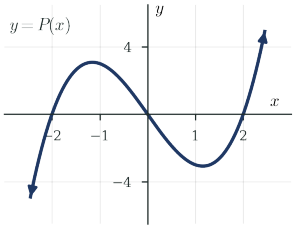
A. $f(x) \to \infty$ as $x \to \infty$ and $f(x) \to -\infty$ as $x \to -\infty$
B. $f(x) \to -\infty$ as $x \to \infty$ and $f(x) \to \infty$ as $x \to -\infty$
C. $f(x) \to -\infty$ as $x \to \infty$ and $f(x) \to -\infty$ as $x \to -\infty$
D. $f(x) \to \infty$ as $x \to \infty$ and $f(x) \to \infty$ as $x \to -\infty$

2 Let $g(x) = -x^2(2x-1)(x+4)$. What is $\lim_{x \to -\infty} g(x)$?

A. ∞ **B.** 0
C. -2 **D.** $-\infty$

3 The graph of a polynomial P is shown. Which could be the degree and leading coefficient of P?

A. Degree 3, positive leading coefficient
B. Degree 3, negative leading coefficient
C. Degree 2, positive leading coefficient
D. Degree 4, negative leading coefficient



4 A polynomial function has degree 6. What is the greatest number of turning points its graph can have?

A. 6 **B.** 7
C. 5 **D.** 3

5 Let $h(x) = 5x - x^3 + 2x^4 - 9$. Which statement is true?

A. $h(x) \to -\infty$ as $x \to -\infty$ and $h(x) \to \infty$ as $x \to \infty$
B. $h(x) \to \infty$ as $x \to -\infty$ and as $x \to \infty$
C. $h(x) \to \infty$ as $x \to -\infty$ and $h(x) \to -\infty$ as $x \to \infty$
D. $h(x) \to -\infty$ as $x \to -\infty$ and as $x \to \infty$

6 A polynomial q has three turning points, and $q(x) \to -\infty$ as $x \to \infty$ and as $x \to -\infty$. Which could be the degree and leading coefficient of q?

A. Degree 4, negative leading coefficient
B. Degree 4, positive leading coefficient
C. Degree 3, negative leading coefficient
D. Degree 5, negative leading coefficient

7 Which function satisfies $f(x) \to \infty$ as $x \to -\infty$ and $f(x) \to -\infty$ as $x \to \infty$?

A. $f(x) = x^3 + 6x$ **B.** $f(x) = -x^4 + 6x$
C. $f(x) = x^4 - 6x$ **D.** $f(x) = -x^3 + 6x$

8 Let $f(x) = (x-2)(x+1)(3-x)$. Which statement is true?

A. $f(x) \to -\infty$ as $x \to -\infty$ and $f(x) \to \infty$ as $x \to \infty$
B. $f(x) \to \infty$ as $x \to -\infty$ and as $x \to \infty$
C. $f(x) \to \infty$ as $x \to -\infty$ and $f(x) \to -\infty$ as $x \to \infty$
D. $f(x) \to -\infty$ as $x \to -\infty$ and as $x \to \infty$

Polynomial Functions and End Behavior

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

At the outdoor music festival, the number of visitors inside the grounds, in hundreds, is modeled by $N(t) = -0.5t^3 + 8t^2 + 2t$, where t is the number of hours after the gates open and $0 \le t \le 16$.

(a) Write $\lim_{t \to \infty} N(t)$. Explain why the model cannot describe the grounds for all $t \ge 0$. DOK3 2 pt

(b) Find the greatest number of visitors predicted on $0 \le t \le 16$ and the time it happens. Give t and $N(t)$ to three decimal places. DOK3 2 pt

(c) Find the first time after opening when the model predicts $N(t) = 0$ again. Give your answer to three decimal places. DOK3 2 pt

Question 10 No calculator · 6 points

Let $r(x) = -x^3(x-2)^2(x+1)$.

(a) State the degree of r and the sign of its leading coefficient. DOK3 2 pt

(b) Write $\lim_{x \to -\infty} r(x)$ and $\lim_{x \to \infty} r(x)$. DOK3 2 pt

(c) Explain why the graph of r has at least one turning point and at most five. DOK3 2 pt